Ej Orr

Software Engineering Intern — Summer 2027

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Education

University of Vermont Burlington, VT

Bachelor of Science in Computer Science *Expected Dec. 2027*

Relevant Coursework: Secure Distributed Computation, Algorithm Design, Data Structures, Programming Languages, Computability & Complexity, Computer Organization

Experience

Team Lead, ORCA Program May 2026 – Present

University of Vermont *Burlington, VT*

- Lead development of an open-source replacement for the UVM Career Center's quiz-taking platform, eliminating a $80K+/year licensing cost and improving integration with Career Center operations
- Designed and built a SQLite-backed API serving 100+ students per day that tracks quiz progress, stores user responses securely, and aggregates results into clearer retention analytics for Career Center staff and academic advisors
- Reviewed and merged teammates' pull requests and ran Agile sprint planning and daily standups across a team of 3 to keep delivery on schedule

Teaching Assistant, Computer Science Department Aug. 2025 – Present

University of Vermont *Burlington, VT*

- Support students 1-on-1 and in office hours across 7 courses spanning Introduction to Advanced Programming
- Helped students debug memory-corruption issues in C++, build intuition for core data structures, and work through automata theory including Turing machines and decidability

Projects

OpenGL Game Engine | *C++, OpenGL, CMake* Ongoing

- Built a 3D game engine from scratch on OpenGL 3.3 core: per-fragment lighting, channel-aware texture loading with mipmapping, procedurally generated maps via randomized walk, and swept AABB collision detection with angle-correct response
- Implemented a data-oriented entity-component system using dense component pools with generation-checked entity handles and O(1) swap-remove, driven by a 60 Hz fixed-timestep simulation loop decoupled from render frame rate
- Built debug tooling (Dear ImGui inspector, wireframe and collider overlays, hot-reloadable shaders) and set up CMake presets, an ASan/UBSan build, CTest unit tests, and GitHub Actions CI
- Originally developed for coursework, then expanded and open-sourced on GitHub for presentation at the UVM CS Fair

UVM Career Quiz | *Python, Flask, SQLite, JavaScript* May 2026 – Present

- Built the Flask backend for the ORCA quiz platform (11 routes) with session-based auth and a SQLite schema modeling per-quiz ownership and role-based sharing/revoking, serving a vanilla JS/HTML frontend
- Collaborated with teammates across feature branches, reviewing and merging pull requests for the quiz builder, sharing/role, and secure-login flows

Cirrocumulus | *Rust, Firecracker, nftables, SQLite* Sept. 2026 – Ongoing

- Building a self-hosted mini cloud that runs every workload in its own jailer-sandboxed Firecracker microVM, managed entirely from a `cirro` terminal CLI
- Implemented crash-safe Node agent state in SQLite, reconciling live VM processes by PID and /proc start time on restart so VMs outlive an agent crash instead of being orphaned
- Wrote the nftables egress policy (accept/drop/masquerade) for each Node's carved VM subnet; drove the whole build test-first behind a GitHub Actions CI gate that boots real Firecracker VMs

Technical Skills

Languages: Python, Java, C, C++, Rust, SQL, HTML/CSS
Frameworks: Flask, OpenGL
Developer Tools: Git, Docker, Ollama Model Deployment, Linux, Bash, Visual Studio, Vim
Exposure: Swift, OCaml, Lua, React, PyTorch, Google Maps SDK